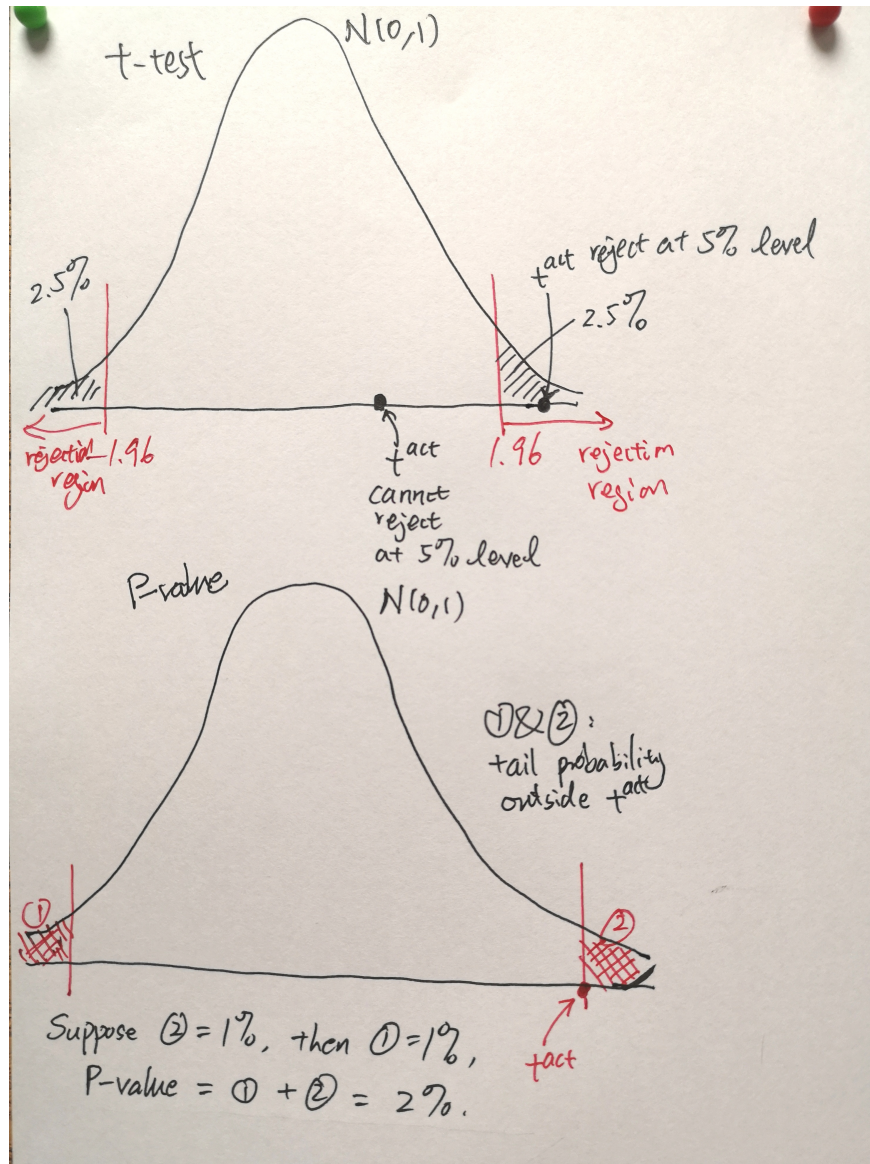


1. (a) See the following figure for the difference between the procedure of t -test (with 5% significance level) and that of p -value (H_0 is rejected at 5% significance level if p -value is smaller than 5%).



- (b) For the linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$, $i = 1, \dots, n$, the 95% confidence interval (CI) for β_1 collect all the possible values of β_1 that cannot be rejected by the

corresponding (two-sided) t -test at the 5% significance level:

$$CI_{\beta_1, 95\%} = \left\{ \beta_1 : \left| \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \right| \leq 1.96 \right\} \quad (1)$$

$$= \left\{ \beta_1 : \left| \hat{\beta}_1 - \beta_1 \right| \leq 1.96 \cdot SE(\hat{\beta}_1) \right\} \quad (2)$$

Solving the inequality, we find that the 95% CI takes the following form:

$$CI_{\beta_1, 95\%} = \left[\hat{\beta}_1 - 1.96 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \cdot SE(\hat{\beta}_1) \right], \quad (3)$$

where $\hat{\beta}_1$ denotes the OLS estimator for β_1 and $SE(\hat{\beta}_1)$ denotes the standard error for $\hat{\beta}_1$. We can define the 95% CI for β_0 in the same fashion.

[Note: according to the formula in (3), the center of the CI is in fact equal to the OLS estimator $\hat{\beta}_1$ (that is, the CI is symmetric around the OLS estimator $\hat{\beta}_1$), and the length of the CI is equal to $2 \cdot 1.96 \cdot SE(\hat{\beta}_1)$. This implies that the smaller the standard error, the shorter the CI will be (this is reasonable because a small standard error means that the estimation for β_1 is precise; by contrast, a large standard error means that the estimation is imprecise).]

To construct a 99% confidence interval, we just need to replace 1.96 with 2.58, which is the critical value for the (two-sided) t -test at the 1% significance level:

$$CI_{\beta_1, 99\%} = \left[\hat{\beta}_1 - 2.58 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 2.58 \cdot SE(\hat{\beta}_1) \right]. \quad (4)$$

By definition, a 99% CI has a higher coverage probability than a 95% CI: it will cover the true value 99% of the time. However, according to the formula in (4), the price to pay for getting a higher coverage probability is that the length of the 99% CI also has to be longer than that of the 95% CI.

- (c) By the definition of a 95% CI for β_1 , it collects all the possible values of β_1 that cannot be rejected by the corresponding (two-sided) t -test at the 5% significance level. Then, suppose the value 1 is inside the 95% CI for β_1 , then it cannot be rejected by the t -test at the 5% level. More specifically, we can conclude that for testing $H_0 : \beta_1 = 1$,

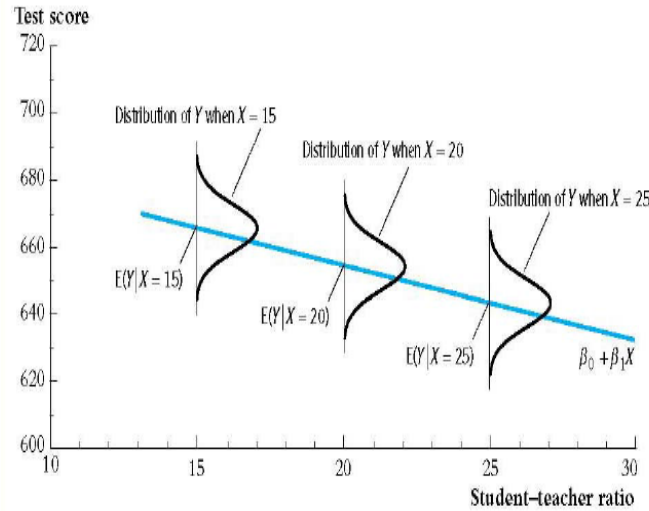
$$|t_1| = \left| \frac{\hat{\beta}_1 - 1}{SE(\hat{\beta}_1)} \right| \leq 1.96, \quad (5)$$

where we use the notation t_1 to emphasize that we are testing $H_0 : \beta_1 = 1$. On the other hand, suppose the value 2 is outside the 95% CI, then we can conclude that for testing $H_0 : \beta_1 = 2$,

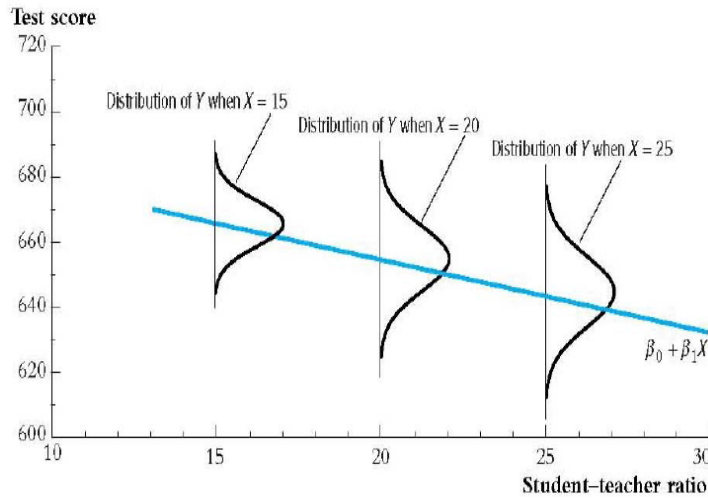
$$|t_2| = \left| \frac{\hat{\beta}_1 - 2}{SE(\hat{\beta}_1)} \right| > 1.96. \quad (6)$$

- (d) Homoskedasticity: In the following figure, the distribution of Y (and the distribution of the errors u) is shown for different values of X (remember that the only difference

between Y and u is the population regression line). All these distributions have the same spread: The variance of these distributions is the same for different values of X . Thus, the errors u in this figure are homoskedastic (the variance of u does not change with the value of X : $Var(u_i|X_i) = c$, where c is a constant).



Heteroskedasticity: By contrast, the following figure illustrates a case in which the distribution of Y (and the distribution of u) spreads out as the value of X increases. In general, the errors u are called heteroskedastic when the variance of u changes with the value of X ($Var(u_i|X_i) = f(X_i)$).



2. (a) By using conditional expectation, we have $\beta_1 = E[TestScore|SmallClass = 1] - E[TestScore|SmallClass = 0]$. As the OLS estimate $\hat{\beta}_1 = 13.9$, on average small classes improve test scores by 13.9 (points).
- (b) To test if the estimated effect of class size on test score is statistically significant, we

compute:

$$|t| = \left| \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \right| = \left| \frac{13.9}{2.5} \right| = 5.56 > 1.96. \quad (7)$$

Therefore, we can reject $H_0 : \beta_1 = 0$ at the 5% level, which means that the estimated effect is statistically significant at the 5% level.

(c) By the definition of a 95% CI, we have:

$$\begin{aligned} CI_{\beta_1, 95\%} &= [\hat{\beta}_1 - 1.96 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \cdot SE(\hat{\beta}_1)] \\ &= [13.9 - 1.96 \cdot 2.5, 13.9 + 1.96 \cdot 2.5] \\ &= [9, 18.8]. \end{aligned} \quad (8)$$

3. (a) For $H_0 : \beta_1 = 5$, the p -value is equal to 0.012, which is larger than 0.01, so we cannot reject $H_0 : \beta_1 = 5$ at the 1% significance level. Since we cannot reject it at the 1% level, by the relationship between t -test and confidence interval, we can conclude that the value $\beta_1 = 5$ will be contained in the 99% confidence interval.

Furthermore, since the p -value is smaller than 0.05, so we can reject $H_0 : \beta_1 = 5$ at the 5% significance level. Since we can reject it at the 5% level, the value $\beta_1 = 5$ will not be contained in the 95% confidence interval.

- (b) To find out the distribution of the t -statistic when the true value of β_1 is equal to 10 instead of 5, note that in general the following always holds under large samples (and when all the OLS assumptions are satisfied):

$$\frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim N(0, 1). \quad (9)$$

Therefore, for testing $H_0 : \beta_1 = 5$,

$$\begin{aligned} t_5 &= \frac{\hat{\beta}_1 - 5}{SE(\hat{\beta}_1)} \\ &= \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} + \frac{\beta_1 - 5}{SE(\hat{\beta}_1)} \\ &\sim N(0, 1) + \frac{10 - 5}{SE(\hat{\beta}_1)} = N(0, 1) + 5 = N(5, 1), \end{aligned} \quad (10)$$

which follows from the information that the true value $\beta_1 = 10$ and $SE(\hat{\beta}_1) = 1$. Therefore, when the true value of β_1 is 10, the distribution of the t -statistic for testing $H_0 : \beta_1 = 5$ actually follows $N(5, 1)$, a mean-shift toward right-hand-side by 5 from the $N(0, 1)$ distribution.

Then, we can draw the graph for the corresponding area of Type II error as follows. Remember the definition of the Type II error is: accepting H_0 when H_0 is wrong (when H_1 is correct), therefore **we just need to find out the area corresponding to the**

H_1 distribution $N(5, 1)$ that falls inside the Acceptance Region (from -1.96 to 1.96). Apparently this area is smaller than 50% (see Figure 1 below).

In general, remember that $N(0, 1)$ is the distribution of the t -test when H_0 is true, $N(c, 1)$ with $c \neq 0$ is the distribution of the t -test when H_0 is wrong.

Similarly, if we consider the case where the H_1 distribution is $N(2, 1)$, we see that now the Type II error becomes much larger as the distribution shifts toward left-hand-side, but it is still slightly smaller than 50%, as 2 is larger than 1.96 (see Figure 2 below).

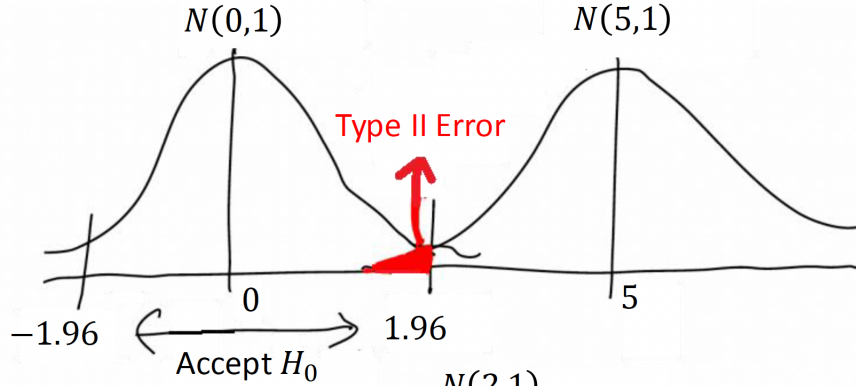


Figure 1

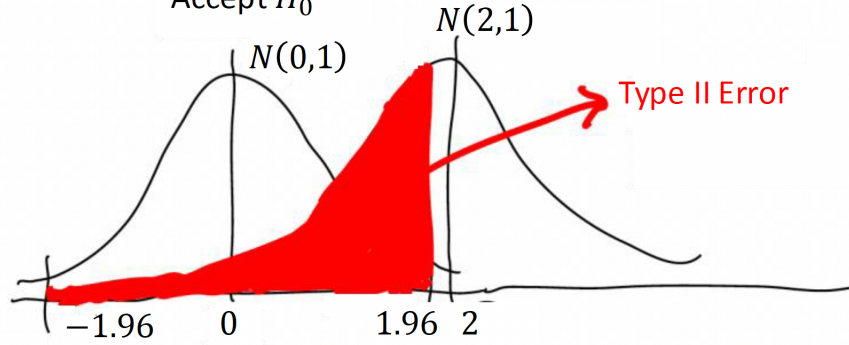


Figure 2

4. (a) For $H_0 : \mu = 10$, the p -value is 0.07, which is smaller than 0.1, so we can reject $H_0 : \mu = 10$ at the 10% significance level. Since we can reject it at the 10% level, then $\mu = 10$ is not contained in the 90% confidence interval.

- (b) For testing $H_0 : \mu = 7$, note that the corresponding t -statistic can be written as:

$$t_7 = \frac{\hat{\mu} - 7}{SE(\hat{\mu})} = \frac{\hat{\mu} - 10 + 10 - 7}{SE(\hat{\mu})} = \frac{\hat{\mu} - 10}{SE(\hat{\mu})} + \frac{3}{SE(\hat{\mu})} = t_{10} + \frac{3}{SE(\hat{\mu})} > t_{10}, \quad (11)$$

which follows from the fact that $SE(\hat{\mu})$ is always positive. Therefore, t_7 is always on the right-hand-side of t_{10} .

Furthermore, we notice that for testing $H_0 : \mu = 10$, the p -value is 0.07, so it cannot be rejected by a (two-sided) t -test at the 5% level. Therefore, t_{10} will be inside the acceptance region from -1.96 to 1.96. **On the other hand, whether t_7 is inside the region from -1.96 to 1.96 depends on the value of $SE(\hat{\mu})$.**

Let us think about some examples. Suppose the standard error $SE(\hat{\mu})$ is very small such as 0.001, then $t_7 = t_{10} + 3000$, so that t_7 must be outside the acceptance region

$[-1.96, 1.96]$. Then, suppose the standard error is very large such as 1000, then $t_7 = t_{10} + 0.003$, so that t_7 is very close to t_{10} . In this case, t_7 will be inside the acceptance region. In sum, depending on the value of the standard error, t_7 can be either inside or outside the acceptance region. So we cannot determine if $H_0 : \mu = 7$ is rejected at the 5% level by the information at hand.